

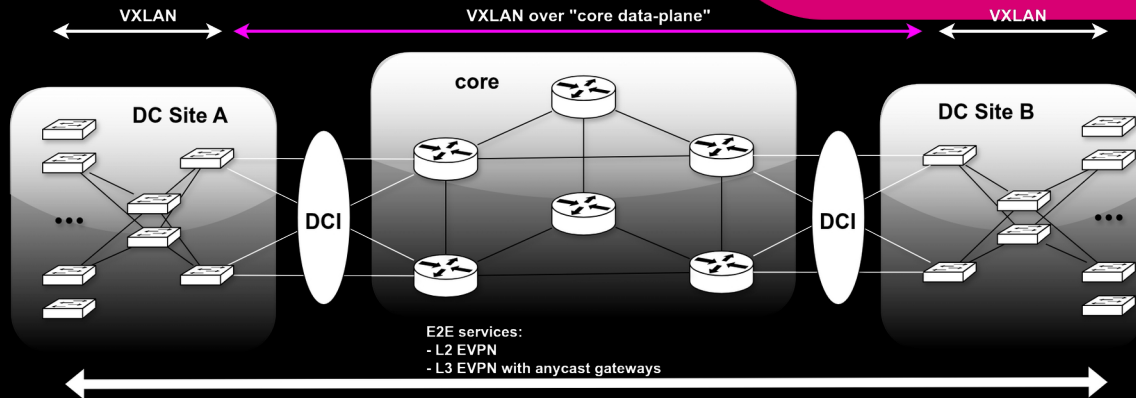
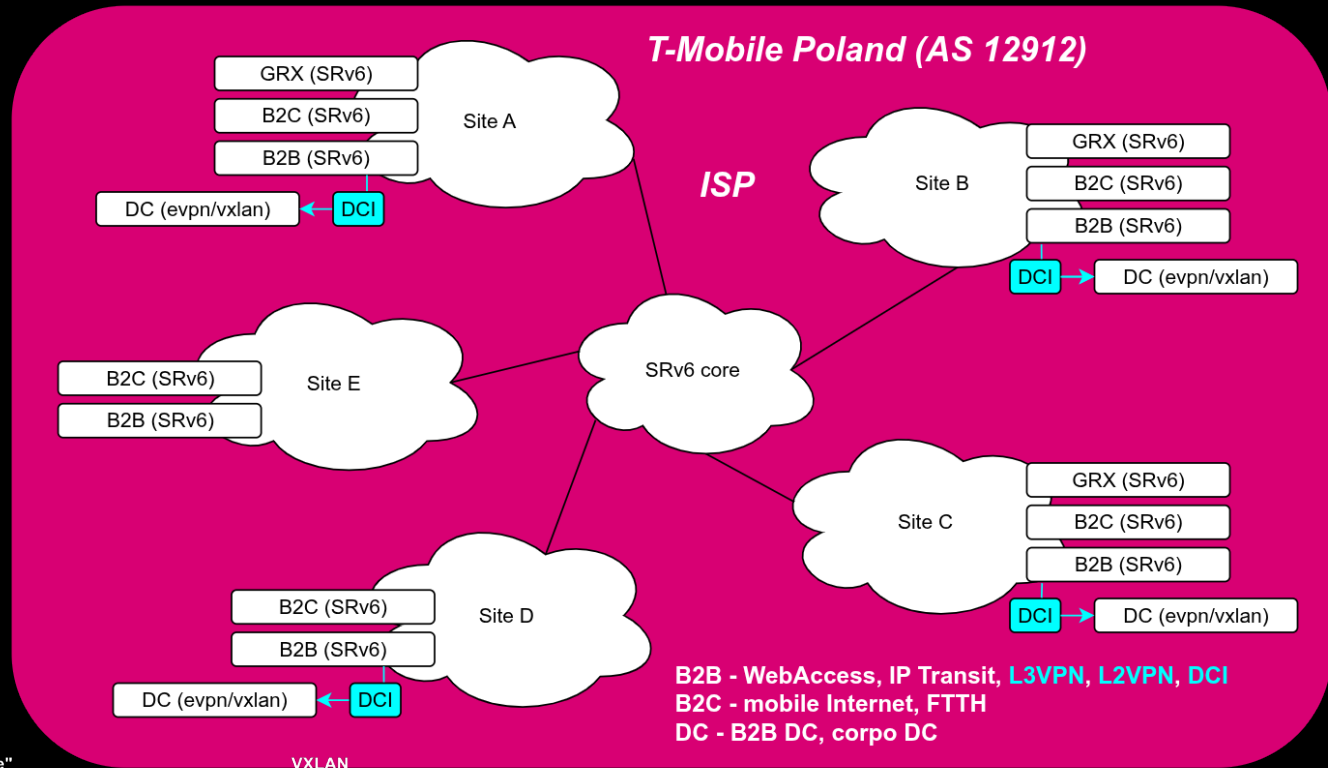


T-Mobile Poland SRv6 deployment



ISP & DC/DCI

- A common core
- Multiple logical domains
- DC and DCI integration enables seamless service extension across the country



SR-MPLS phase (and why it didn't survive)

WHY?

Introduced around 2021 in the GRX domain to meet the dual-plane requirement for signaling traffic.

PROS

- Fantastic simplification of traffic engineering compared to RSVP-TE.
- Dynamic traffic-engineering tunnel creation using ODN and auto-steering.
- Smooth and straightforward migration with no major issues.

CONS

- Ultimately, it shares the same limitations as MPLS.
- Not a future-proof solution.
- Unable to provide true unification across network domains.



SRv6 deployment considerations

- Which hardware platforms support SRv6, and how should we handle SRv6-unaware devices?
- How to prepare infrastructure(sw upgrade, hw swap)?
- How will existing services be affected?
 - Suboptimal traffic forwarding
 - How will multi-homed customers be handled?
 - MTU considerations
 - Can all existing services be migrated to SRv6?
- Are all required features supported?
- How should the infrastructure be secured?
- Technology standardization



SRv6 deployment

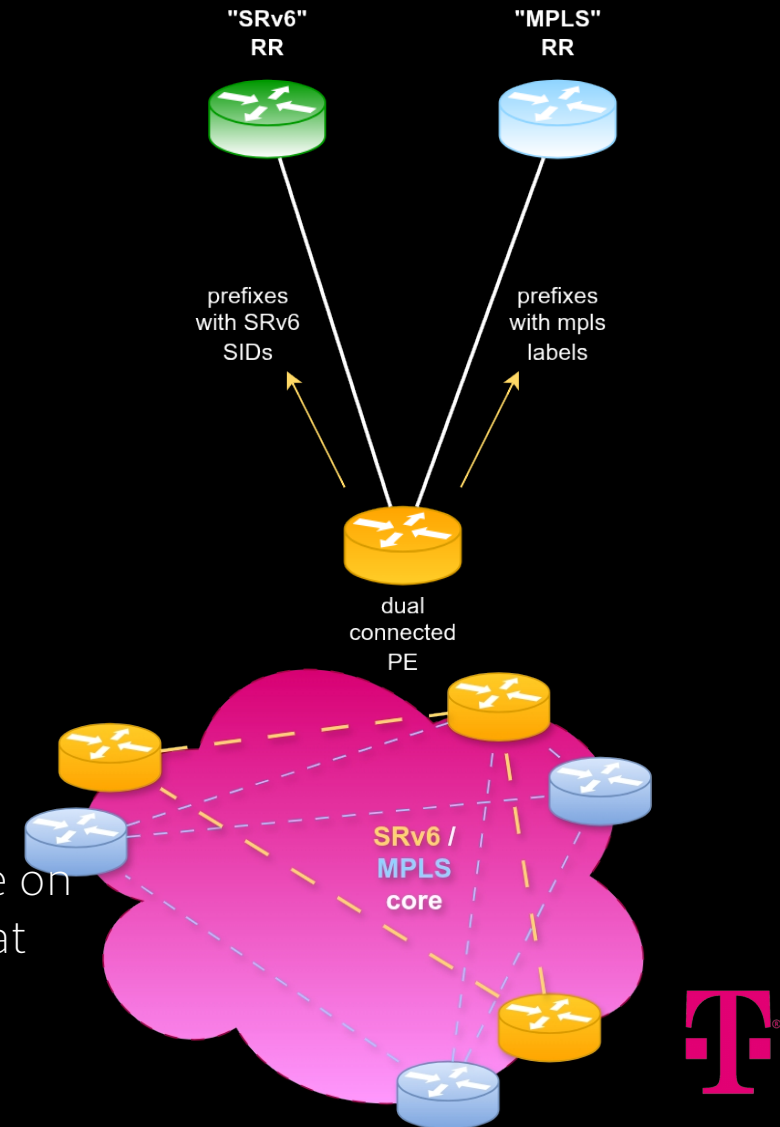


- The entire preparation and migration process was developed and executed in-house.
- A multi-vendor network environment was a key assumption from the very beginning.
- Automation was leveraged throughout the migration process.
- Scope of the deployment: whole ISP network with all service types.

Service migration approach

Dual-Connected PE Approach

- Establish iBGP sessions to the new route reflectors.
- Operate two parallel data planes simultaneously.
- Per router (PE) migration.
- **Slightly** prefer SRv6 data-plane.
- Each PE connected to the SRv6 domain will immediately prefer SRv6 paths.
- We additionally developed a new L2EVPN (P2MP) service on our SRv6 network, so we do not migrate any L2 services at all.



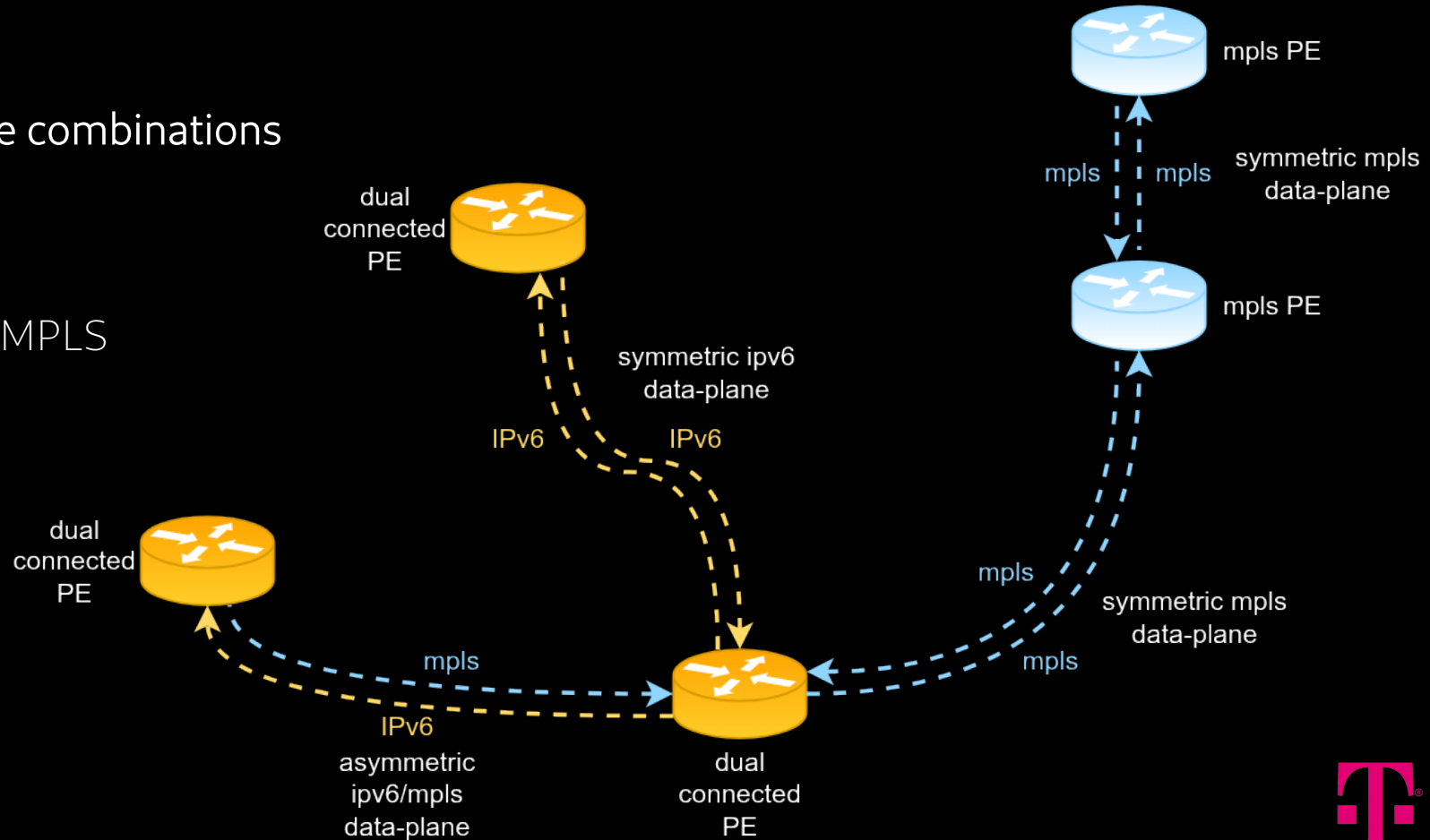
Data-planes coexistence

Data-planes possible combinations

- symmetric MPLS
- symmetric SRv6
- asymmetric SRv6/MPLS

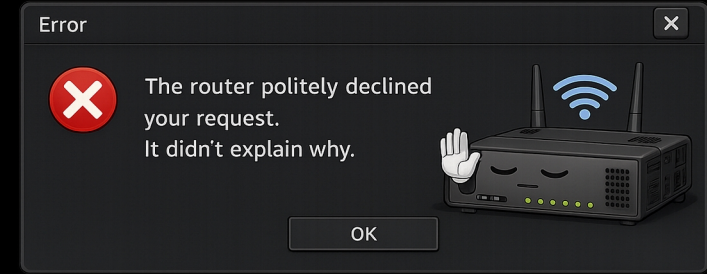


**It worked
flawlessly!**



Challenges we encountered

- Internet services: Netflow
- Global: SRv6 fwd on some hardware with feature combination
- VPN services: MPLS/SRv6 gateway quirks in the non-basic VRF leaking scenarios
- Some TE features were not yet available
- Inter-vendor interoperability required extensive validation
- Minor operational limitations remain after the migration



SRv6 Operational & Security Readiness

- OSS support
 - Analytical & management systems
- Security
 - ACL + secure control-plane
 - BGP attributes filtering
- Network functions
 - Anti-DDoS system



SRv6 deployment summary

The overall perception of the technology is very positive.

- It has proven to be stable.
- It is interoperable across different vendors.
- We are not experiencing any **significant** issues.
- The migration, apart from software and hardware limitations, did not reveal any technology-related drawbacks.
- New features are being developed rapidly.



TMPL One common SRv6 network

~65k B2B services of all types

~450k residential services

~8M mobile customers

Up to 3.5Tbps of daily traffic in peaks



SRv6 – what's next?

- Focus on the data center — replace VXLAN with SRv6 (long-term solution)
- Adopt a containerized platform with SRv6 support
- Introduce SRv6 into the aggregation network
- Evaluate additional features if they provide tangible value
- Network orchestration in the future



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T-Mobile Poland



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